

Course Seven

Google Advanced Data Analytics Capstone



Instructions

Use this PACE strategy document to record your decisions and reflections as a data professional as you work through the capstone project. As a reminder, this document is a resource guide that you can reference in the future and a space to help guide your responses and reflections posed at various points throughout the project.

Portfolio Project Recap

Many of the goals you accomplished in your individual course portfolio projects are incorporated into the Advanced Data Analytics capstone project including:

- Create a project proposal
- Demonstrate understanding of the form and function of Python
- Show how data professionals leverage Python to load, explore, extract, and organize information through custom functions
- Demonstrate understanding of how to organize and analyze a dataset to find the “story”
- Create a Jupyter notebook for exploratory data analysis (EDA)
- Create visualization(s) using Tableau
- Use Python to compute descriptive statistics and conduct a hypothesis test
- Build a multiple linear regression model with ANOVA testing
- Evaluate the model
- Demonstrate the ability to use a notebook environment to create a series of machine learning models on a dataset to solve a problem
- Articulate findings in an executive summary for external stakeholders

Project proposal

Salifort Motors: Employee Attrition — Project Proposal

Overview

Salifort Motors has a high employee turnover rate: 16.6% of employees in the dataset left the company. The goal of this project is to use HR survey data to identify what drives attrition, build a model that predicts whether an employee will leave, and give leadership actionable retention recommendations. The project follows the PACE framework: Plan (problem framing and data), Analyze (EDA and hypothesis tests), Construct (logistic regression, decision tree, random forest, XGBoost), and Execute (evaluation, interpretation, and an executive summary).

Milestones	Tasks	PACE stages
1. Frame the problem	Define audience, goal, and success metrics; review the data dictionary	Plan
2. Clean the data	Rename columns, remove 3,008 duplicates, check outliers and class balance	Plan / Analyze
3. EDA	Visualizations, segmentation of leavers, correlations	Analyze
4. Test hypotheses	Chi-square (salary, department); Mann–Whitney (hours, satisfaction)	Analyze
5. Build models	Logistic regression, decision tree, random forest, XGBoost; 5-fold CV; two feature rounds	Construct



6. Evaluate	Hold-out test, threshold tuning, permutation importance, SHAP, risk scoring	Construct / Execute
7. Communi- cate	Executive summary for leadership, recommendations, pilot plan	Execute

Data Project Questions & Considerations



PACE: Plan Stage

Foundations of data science

- Who is your audience for this project?
Salifort Motors leadership (decisions on retention policy) and the HR department (day-to-day use of the model). Secondary audience: department managers.
- What are you trying to solve or accomplish? And, what do you anticipate the impact of this work will be on the larger business need?
Predict whether an employee will leave and identify the factors behind attrition. Expected impact: lower turnover (currently 16.6%), savings on hiring and training, and higher satisfaction among current employees.
- What questions need to be asked or answered?
Which factors are most strongly associated with leaving? Are there typical profiles of employees who leave? Can we reliably predict attrition before it happens? Which management levers (workload, promotions, pay) are actually available?
- What resources are required to complete this project?
The HR survey dataset (14,999 rows, 10 columns); Python (pandas, scikit-learn, XGBoost, SHAP, SciPy); Jupyter Notebook; scikit-learn and XGBoost documentation; input from HR on what each field means.
- What are the deliverables that will need to be created over the course of this project?
A Jupyter notebook with EDA, hypothesis tests, and models; this PACE strategy document; and a one-page executive summary for leadership.

Get Started with Python

- How can you best prepare to understand and organize the provided information?
Review the data dictionary; check data types, missing values, duplicates, and outliers; standardize column names (fix the average_montly_hours typo, Department → department).
- What follow-along and self-review codebooks will help you perform this work?
Reference guides for pandas (cleaning, groupby, crosstab), matplotlib/seaborn (visualization), and scikit-learn (Pipeline, GridSearchCV, classification metrics), plus my own course notes on PACE.

- What are a couple additional activities a resourceful learner would perform before starting to code?

Ask HR how and when satisfaction was measured, and whether 'left' includes both resignations and terminations; write down hypotheses before coding so conclusions aren't fitted to the data.

Go Beyond the Numbers: Translate Data into Insights

- What are the data columns and variables and which ones are most relevant to your deliverable?
satisfaction_level, last_evaluation, number_project, average_monthly_hours, tenure, work_accident, left (target), promotion_last_5years, department, salary. Most relevant: workload (projects, hours), satisfaction, evaluation, and tenure.
- What units are your variables in?
Satisfaction and evaluation are proportions from 0 to 1; hours are average hours per month; tenure is in years; projects are counts; binary fields are 0/1; salary is a category (low/medium/high); department is a category.
- What are your initial presumptions about the data that can inform your EDA, knowing you will need to confirm or deny with your future findings?
Hypotheses: overwork and low satisfaction drive attrition; low pay and a lack of promotions raise risk; department has a weak effect. All of these were tested in EDA.
- Is there any missing or incomplete data?
No missing values. However, 3,008 rows (20%) are exact duplicates; they were removed, leaving 11,991 rows.
- Are all pieces of this dataset in the same format?
Yes. Numeric fields are stored as numbers; the two categorical fields (department, salary) are strings and were encoded (salary as ordinal, department as one-hot).
- Which EDA practices will be required to begin this project?
Structure and duplicate checks, descriptive statistics, outlier detection (IQR for tenure), class balance (83/17), distributions and relationships with the target, and a correlation matrix.

The Power of Statistics

- What is the main purpose of this project?
Build a model that predicts employee attrition and surface the key drivers leadership can act on.
- What is your research question for this project?

Which employee characteristics (workload, tenure, evaluation, pay, promotions, department) predict whether an employee will leave Salifort Motors?

- What is the importance of random sampling? In this case, what is an example of sampling bias that might occur if you didn't use random sampling?

Random sampling makes the findings representative of the whole company. Example of sampling bias: if only volunteers or a single department took the survey, dissatisfied or overworked employees could be over- or under-represented.

Regression Analysis: Simplify Complex Data Relationships

- Who are your stakeholders for this project?
Salifort Motors leadership, the HR department, and department managers; indirectly, the employees themselves.
- What are you trying to solve or accomplish?
A classification problem: predict the binary variable 'left' and understand the direction of each feature's effect.
- What are your initial observations when you explore the data?
A U-shaped relationship between attrition and project count (2 projects: 54%; 7 projects: 100%); a median of 200 hours/month vs. a ~167-hour standard; attrition peaking in years 4–5; only 1.7% promoted.
- What resources do you find yourself using as you complete this stage? (Make sure to include the links.)
Documentation: scikit-learn (<https://scikit-learn.org/stable/>), XGBoost (<https://xgboost.readthedocs.io/>), pandas (<https://pandas.pydata.org/docs/>), SHAP (<https://shap.readthedocs.io/>).
- Do you have any ethical considerations in this stage?

This is people data: analysis stays aggregated, and the model must not be used for penalties. Satisfaction is a subjective survey score and should be used with care.

The Nuts and Bolts of Machine Learning

- What am I trying to solve?
Binary classification: will an employee leave (left = 1)?
- What resources do you find yourself using as you complete this stage?

scikit-learn and XGBoost documentation, course materials on decision trees and ensembles, and articles on model interpretation (SHAP, permutation importance).

- Is my data reliable?

Mostly yes: no missing values and all values are within valid ranges. But 20% of rows were duplicates, the timing of the survey relative to departure is unclear, and voluntary and involuntary departures aren't separated — these are limitations.

- Do you have any additional ethical considerations in this stage?

Leakage risk: satisfaction may have been recorded after the decision to leave was made, so a second model was built without `satisfaction_level`. Predictions must not be used in a discriminatory way.

- What data do I need/would I like to see in a perfect world to answer this question?

Reason for leaving (voluntary vs. involuntary), date of last raise, manager changes, regular pulse-survey results, time off (aggregated), work arrangement, and commute distance.

- What data do I have/can I get?

The current 10-feature dataset; in production, monthly HRIS extracts (projects, hours, tenure, evaluation, pay).

- What metric should I use to evaluate success of my business objective? Why?

F1 as the primary metric (balances precision and recall under class imbalance), plus recall (missing a departing employee costs more than an extra conversation) and AUC (quality of risk ranking). Accuracy is a poor primary metric: predicting 'nobody leaves' would already score 83%.

Data Project Questions & Considerations



PACE: Analyze Stage

Get Started with Python

- Will the available information be sufficient to achieve the goal based on your intuition and the analysis of the variables?

Yes, for prediction: workload, tenure, and evaluation carry a strong signal. For a deeper understanding of causes, we lack the reason for leaving and salary history.

Go Beyond the Numbers: Translate Data into Insights

- What steps need to be taken to perform EDA in the most effective way to achieve the project goal?

Remove duplicates; check outliers; examine distributions; compare leavers and stayers on each feature; build joint plots (hours × satisfaction, projects × hours); identify segments.

- Do you need to add more data using the EDA practice of joining? What type of structuring needs to be done to this dataset, such as filtering, sorting, etc.?

No joins were needed. Structuring included renaming columns, removing duplicates, filtering tenure outliers for logistic regression, and grouping and cross-tabulation.

- What initial assumptions do you have about the types of visualizations that might best be suited for the intended audience?

For leadership: simple bar charts with labeled percentages and one scatter plot with the three profiles annotated. Heatmaps and SHAP plots are for a technical audience.

The Power of Statistics

- Why are descriptive statistics useful?

They quickly show central tendency, spread, and anomalies (e.g., a median of 200 hours/month vs. a 167-hour standard) and frame the tests and models that follow.

- What is the difference between the null hypothesis and the alternative hypothesis?

The null hypothesis states there is no effect or association (e.g., salary and attrition are independent); the alternative states that there is one. The test measures how incompatible the data is with H_0 .

Regression Analysis: Simplify Complex Data Relationships

- What are some purposes of EDA before constructing a multiple linear regression model?

To detect outliers, multicollinearity, non-linearity, class imbalance, and the need to encode categories — all of which shape the choice and setup of the model (here, logistic regression).

- Do you have any ethical considerations in this stage?

Avoid drawing conclusions about individuals from aggregates, and avoid treating associations as causal without testing.

The Nuts and Bolts of Machine Learning

- What am I trying to solve? Does it still work? Does the plan need revising?

The goal stayed the same. The plan was revised: tree-based models were added because of non-linearity, and a second round without survey data was added because of leakage risk.

- Does the data break the assumptions of the model? Is that ok, or unacceptable?

For logistic regression, partly: the non-linear (U-shaped) relationships violate the linear-logit assumption, and tenure outliers were excluded. That's acceptable for interpreting direction, but not for a production forecast (recall 0.25). Tree-based models don't require these assumptions.

- Why did you select the X variables you did?

All original features, since EDA showed nearly every one is related to attrition. In Round 2, `satisfaction_level` was dropped (leakage risk, dependence on the survey) and `overworked (>175 h/month)` and `hours_per_project` were added.

- What are some purposes of EDA before constructing a model?

To understand the data structure, catch leakage and errors, engineer features, and choose an appropriate model class and metrics.

- What has the EDA told you?

Attrition is explained by three profiles (burnout, underloaded, overworked high performers), covering 93% of leavers; workload and tenure are the key factors; department is not a driver.

- What resources do you find yourself using as you complete this stage?

`seaborn`, `scikit-learn`, and `SciPy` (statistical tests) documentation, plus course materials.

- Do you have any ethical considerations in this stage?



Segments describe groups, not specific people; the 'underloaded' segment may include involuntary terminations and should be interpreted carefully.

Data Project Questions & Considerations



PACE: Construct Stage

Get Started with Python

- Do any data variables averages look unusual?

Yes: average hours (≈ 200 /month) are well above the standard (≈ 167) — overwork is systemic. The promotion rate (1.7%) is unusually low.

- How many vendors, organizations or groupings are included in this total data?

10 departments: sales, technical, support, IT, R&D, product management, marketing, accounting, HR, and management; plus 3 salary levels.

Go Beyond the Numbers: Translate Data into Insights

- What data visualizations, machine learning algorithms, or other data outputs will need to be built in order to complete the project goals?

EDA charts (bar, box plot, scatter, heatmap); logistic regression, decision tree, random forest, and XGBoost; confusion matrix; ROC and precision–recall curves; permutation importance; SHAP; and a surrogate rule tree.

- What processes need to be performed in order to build the necessary data visualizations?

Cleaning and encoding the data, aggregating by the target variable, labeling segments, and annotating charts for a non-technical audience.

- Which variables are most applicable for the visualizations in this data project?

number_project, average_monthly_hours, satisfaction_level, last_evaluation, tenure, promotion_last_5years, salary.

- Going back to the Plan stage, how do you plan to deal with the missing data (if any)?

There were no missing values; instead, duplicates were removed and tenure outliers were excluded only for logistic regression.

The Power of Statistics

- How did you formulate your null hypothesis and alternative hypothesis?

H_0 : salary level and attrition are independent; H_1 : they are associated (chi-square test). H_0 : the distributions of hours and satisfaction are the same for leavers and stayers; H_1 : they differ (Mann–Whitney U test, because the distributions aren't normal). $\alpha = 0.05$.

- What conclusion can be drawn from the hypothesis test?

All null hypotheses were rejected. Salary: $\chi^2 = 175$, $p < 0.001$, Cramér's $V = 0.12$. Hours: median 226 vs. 198, $p < 0.001$. Satisfaction: 0.41 vs. 0.69, $p < 0.001$. Department is formally significant ($p = 0.013$), but the effect is negligible ($V = 0.04$).

Regression Analysis: Simplify Complex Data Relationships

- Do you notice anything odd?

The number_project coefficient in the logistic regression is < 1 ('more projects, less attrition'), which contradicts the EDA. The cause is the U-shaped relationship that a linear model cannot capture.

- Can you improve it? Is there anything you would change about the model?

Yes: move to tree-based models, which naturally capture non-linearity and interactions (or encode project count as a categorical variable in the regression).

The Nuts and Bolts of Machine Learning

- Is there a problem? Can it be fixed? If so, how?

Two issues: non-linearity (solved with tree-based models) and possible leakage through satisfaction_level (solved with the Round 2 model without survey data). Overfitting was controlled with cross-validation and a hold-out set.

- Which independent variables did you choose for the model, and why?

Round 2: number_project, average_monthly_hours, last_evaluation, tenure, work_accident, promotion_last_5years, salary, department, overworked, hours_per_project — all available in the HR system without a survey.

- How well does your model fit the data? (What is my model's validation score?)

5-fold CV for the Round 2 random forest: $F1 = 0.922$, $AUC = 0.975$. On the hold-out set: precision 0.95, recall 0.92, $F1 = 0.935$, $AUC = 0.97$, accuracy 0.98.

- Can you improve it? Is there anything you would change about the model?

Expand the hyperparameter grid, calibrate probabilities, add new features (salary history, manager changes), and separate voluntary from involuntary departures.



- Do you have any ethical considerations in this stage?



High risk is a reason to offer support, not to penalize. Department and salary level must not become grounds for discrimination; model errors should be audited by group on a regular basis.

Data Project Questions & Considerations



PACE: Execute Stage

Get Started with Python

- Given your current knowledge of the data, what would you initially recommend to your manager to investigate further prior to performing an exploratory data analysis?

Clarify where the duplicates came from, the date and method of the satisfaction survey, and how 'left' is defined (voluntary vs. involuntary).

- What data initially presents as containing anomalies?

Duplicates (20% of rows), tenure outliers (7–10 years), extreme hours (up to 310/month), and 100% attrition at 7 projects.

- What additional types of data could strengthen this dataset?

Reason for leaving, date of last raise, manager changes, exit-interview results, regular pulse surveys, time off, and work arrangement.

Go Beyond the Numbers: Translate Data into Insights

- What key insights emerged from your EDA and visualizations(s)?

Workload is the main driver: the sweet spot is 3–4 projects (1–6% attrition), while 6–7 projects mean 45–100%. Three profiles explain 93% of departures. The risk window is years 4–5. Promotions are almost nonexistent. Department is not a cause.

- What business recommendations do you propose based on the visualization(s) built?

Cap workload at 4–5 projects; monitor anyone above 200 hours/month; pay for overtime or give comp time; run career and pay reviews in years 3–5; evaluate results, not hours.

- Given what you know about the data and the visualizations you were using, what other questions could you research for the team?

Why do satisfied high performers (profile 3) leave? Which 'underloaded' employees were let go? How do manager changes affect attrition? What does a departure cost by role?

- How might you share these visualizations with different audiences?

Leadership: a one-page summary with two charts. HR: the notebook and a dashboard with the at-risk list. Technical peers: the notebook with code and metrics.

The Power of Statistics

- What key business insight(s) emerged from your A/B test?

No A/B test was run in this project; chi-square and Mann–Whitney tests were used instead. I recommend an A/B pilot of retention measures (e.g., a project cap) in 1–2 departments to test the causal effect.

- What business recommendations do you propose based on your results?

Roll out workload and career-growth measures through a pilot with a control group and measure the change in attrition over 2–3 quarters.

Regression Analysis: Simplify Complex Data Relationships

- To interpret model results, why is it important to interpret the beta coefficients?

Coefficients (and odds ratios) show the direction and strength of each feature's association, holding the others constant — which makes the model explainable to the business.

- What potential recommendations would you make to your manager/company?

Satisfaction is the strongest protective factor ($\times 0.34$ odds per 1 SD); tenure is the strongest risk factor ($\times 2.6$). I recommend regular satisfaction monitoring and mid-tenure retention programs.

- Do you think your model could be improved? Why or why not? How?

Yes: the linear model misses 3 of every 4 leavers (recall 0.25) because of non-linear relationships; improvements include non-linear feature transformations or tree-based models.

- What business recommendations do you propose based on the models built?

The regression served as an interpretive baseline; business recommendations rest on the consistent findings from EDA, statistical tests, and tree-based models.

- What key insights emerged from your model(s)?

The direction of effects matches the EDA for satisfaction, tenure, and salary, but not for project count (because of non-linearity).

- Do you have any ethical considerations at this stage?

Do not treat associations as causal, and do not use the model to assess individual employees without human review.

The Nuts and Bolts of Machine Learning

- What key insights emerged from your model(s)?

The key predictors are number of projects, last evaluation, tenure, and hours. The model flags ~180 current employees (1.8%) as at risk; the typical profile is 5 projects, ~250 hours/month, 5 years of tenure, and an evaluation of 0.87.

- What are the criteria for model selection?

Mean F1 from 5-fold cross-validation (primary), plus recall and AUC; practicality in production (no dependence on the survey); and interpretability.

- Does my model make sense? Are my final results acceptable?

Yes: the model's findings match the EDA and statistics, and hold-out performance is close to CV (no overfitting). The results are acceptable for a pilot rollout.

- Were there any features that were not important at all? What if you take them out?

Department, work accidents, and promotions barely affect predictions (promotions because they are so rare). Removing them hardly changes performance, but department is kept for department-level reporting.

- Given what you know about the data and the models you were using, what other questions could you address for the team?

Which retention measures work best? How much would lower turnover save? How quickly does an employee move from low to high risk?

- What resources do you find yourself using as you complete this stage?

scikit-learn, XGBoost, and SHAP documentation; course materials on evaluating classification models.

- Is my model ethical?

The model is ethical as long as it is used only to support employees, is transparent to HR, is regularly audited for errors by group, and the data is stored with restricted access.

- When my model makes a mistake, what is happening? How does that translate to my use case?

A false negative (8% of leavers missed) means the company loses the chance to retain someone. A false positive (5% of flags) means an unnecessary conversation about workload — harmless in a supportive format. That's why the threshold can be lowered to 0.4 for more coverage.